

Selected Projects & Technical Skills

Yuan Gao | Thermal / AI hardware liquid-cooling engineering

Thermal-Hydraulic Co-Design of a 6 kW AI Accelerator Cooling Tray

Python • COMSOL Multiphysics • Reduced-order models • Liquid-cooling networks

- Developed a scalable Python thermal-hydraulic model of an 8-accelerator, 6 kW liquid-cooling tray, integrating CFD-derived cold-plate ROMs, manifolds, branch hardware losses, and a representative pump curve.
- Calibrated physics-based cold-plate ROMs over 0.50–1.20 L/min; reproduced a withheld COMSOL case within 0.25% thermal-resistance error and 0.04°C maximum-base-temperature error.
- Evaluated header sizing and six operating/fault scenarios; predicted 72.7°C nominal peak device temperature with 12.3°C margin, and identified 18.3% flow maldistribution under a 30% branch-resistance increase before exceeding the temperature limit.

Physics-Based & ML-Assisted Microchannel Cold-Plate Optimization

COMSOL Multiphysics • Python • Gaussian process regression • Pareto analysis

- Built a 3D conjugate heat-transfer model for a 100 W, 10-channel liquid cold plate and developed a Gaussian-process surrogate from a 45-case flow/TIM design dataset.
- Used the surrogate to explore thermal-hydraulic tradeoffs and reran three representative Pareto candidates in COMSOL for high-fidelity confirmation.
- Selected a balanced 0.188 L/min candidate with 64.73°C maximum chip temperature; reduced hydraulic pumping power by 63.5% versus the minimum-temperature candidate while retaining 20.27°C temperature margin.

Technical Skills

Thermal & Liquid Cooling: Direct-to-chip cooling; cold plates and microchannels; conjugate heat transfer; thermal-resistance budgeting; lumped TIM/package modeling; thermal design margins.

CFD & System Modeling: COMSOL Multiphysics; parametric CFD; physics-based ROMs; 0D/1D hydraulic networks; manifold flow distribution; Darcy–Weisbach/local-loss models; pump–system operating points.

Programming & Verification: Python; NumPy; SciPy; pandas; Matplotlib; nonlinear solvers; pytest; conservation/residual checks; data visualization; Git/GitHub.

ML & Optimization: Gaussian process regression; design of experiments; cross-validation; held-out CFD comparison; Pareto tradeoff analysis; sensitivity and fault-scenario studies.